Detecting Malignant Skin Lesions from Image and Clinical Data

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This solo project classified skin lesions as needing treatment or not, combining smartphone images with patient clinical metadata from PAD-UFES-20 [2]. Much clinical metadata was missing (patients skipped fields, or curators removed unverifiable entries), so handling missingness well was a core part of the task. The project had two parts: build a random forest (RF) model on engineered image features, then design a new model that outperforms it on held-out test data.

RF Model

I engineered three image features: mean color intensity and color variance (all channels/pixels, capturing pigmentation and color irregularity) and edge density (grayscale gradient-magnitude thresholding, capturing border complexity), merged with the tabular data by `img_id`. Numeric fields were median-imputed; binary fields were normalized to true/false/missing and one-hot encoded; other missing values were coded as an explicit "Missing" category rather than dropped, since absence itself was informative (unverified $\neq$ negative). The training set was heavily imbalanced (BCC alone was 57%), so I augmented (flip, rotate, mild brightness/contrast) all classes except BCC, scaling copies to each class's rarity (e.g., 6× for MEL/SEK, 0× for BCC) to boost minority representation without fully equalizing class sizes.

Cross-validation used `StratifiedGroupKFold` on patient ID to prevent leakage. On a held-out split, I examined per-class decision thresholds ($\tau = 0.1$ for MEL, $\tau = 0.25$ for NEV) to understand the precision/recall trade-off, since missing a malignant MEL is far costlier than over-flagging a benign NEV. I then ran a 5-fold hyperparameter search optimizing *minimum age-group* macro AUROC, a threshold-free metric, so the model couldn't win by only fitting the dominant class or age group. The final model (`n_estimators`=600, `max_depth`=15, `max_features="sqrt"`, `min_samples_split`=12, `min_samples_leaf`=5) reached a test AUROC of 0.901, with high recall (0.996) but low specificity (0.304): it rarely misses a dangerous lesion but produces more false alarms, an acceptable trade-off for a treatment-triage tool.

Open-Ended Challenge

For Part 2, I moved to a fusion model: a frozen, pretrained EfficientNet-B0 (small parameter count, to limit overfitting on a small dataset) producing a 256-d image embedding, concatenated with an MLP embedding of the tabular features and fed to a classification head. I also re-engineered the image features using the dermatology ABCDE framework [1]: grayscale color entropy (replacing a single mean-color scalar), per-channel RGB variance, and diameter-based features (mean diameter, diameter >6mm). An ablation study showed diameter features paired with edge density gave the best minimum age-group AUROC, so the final feature set was diameter mean, diameter >6mm, and edge density.

I also improved sampling: instead of shuffled fine-label batches, I used a `WeightedRandomSampler` (with replacement) over coarse labels to oversample rare age×class combinations, additionally oversampling the negative senior group 1.5× after the model over-predicted negative there. This raised mean minimum age-group AUROC from 0.803 to 0.856 ($\pm$0.066) across folds (power=0.15). Early stopping (patience=3) prevented overfitting past peak validation. A configuration with a higher mean validation AUROC (0.871) was rejected for its larger fold-to-fold variance ($\pm$0.085 vs. $\pm$ 0.066); the final model (lr=0.001, weight decay=0.0001, dropout=0.3, batch size=16, power=0.2) reached a test AUROC of 0.863 and minimum age-group AUROC of 0.797. The fusion model had a lower AUROC than the RF, but better balanced accuracy (0.793 vs. 0.650).

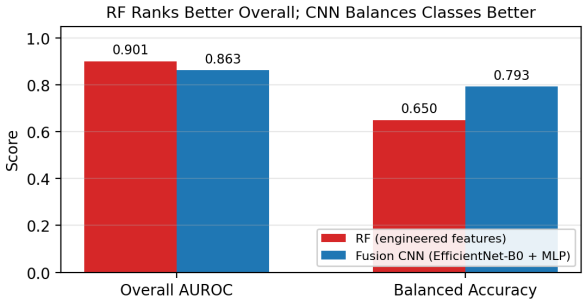


Figure 1: RF ranks lesions needing treatment better overall, but the CNN's balanced accuracy is substantially higher, thanks to weighted sampling of underrepresented subgroups.

Takeaways & Next Steps



Figure 2: Random forest: top 3 confidently-correct and incorrectly classified held-out predictions.

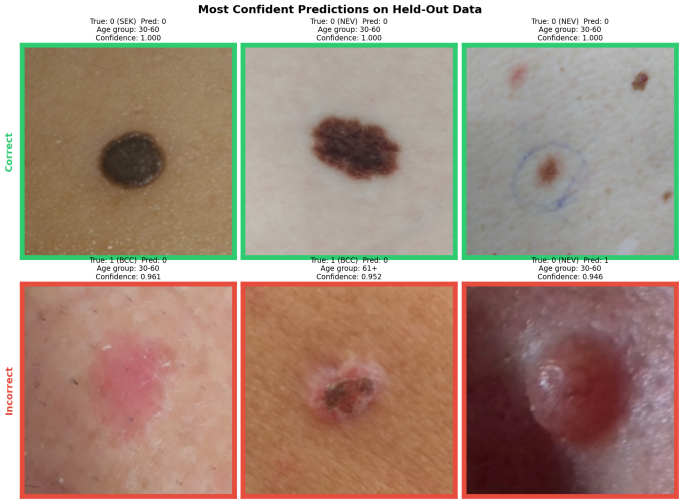


Figure 3: Fusion CNN: top 3 confidently-correct and incorrectly classified held-out predictions.

From these images we can see that these two models are complimentary to each other. The RF model was better at predicting malignant lesions while the fusion model was better at predicting benign lesions. Note that the 2/6 images in 2 have blue surgical marks and the RF model classify them both as malignant, two things are possible: because of the blue ink this affected edge density enough to affect our prediction or the model learned that blue ink means biopsy (surgical marks are a known false-positive source in classifiers). In the fusion model, one of the images had a blue surgical mark but was correctly classified as benign, so the fusion model did not learn that pattern as strongly as the RF model. Instead, decisions were based on color, contrast, and texture, which is consistent with CNNS. The next step would be either an ensemble (like averaging the two probabilities) or weigh the clinical features more in the fusion model (like balancing the two branches better).

References

[1] American Academy of Dermatology. What to look for: ABCDEs of melanoma. `https://www.aad.org/public/diseases/skin-cancer/find/at-risk/abcdes`, 2026. Accessed July 4, 2026.

[2] Andre G.C. Pacheco, Gustavo R. Lima, Amanda S. Salomão, Breno Krohling, Igor P. Biral, Gabriel G. de Angelo, et al. Pad-ufes-20: A skin lesion dataset composed of patient data and clinical images collected from smartphones. *Data in Brief*, 32:106221, 2020.